

The balancing trick: Optimized sampling of imbalanced datasets—A brief survey of the recent State of the Art

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Abstract

This survey paper focuses on one of the current primary issues challenging data mining researchers experimenting on real-world datasets. The problem is that of imbalanced class distribution that generates a bias toward the majority class due to insufficient training samples from the minority class. The current machine learning and deep learning algorithms are trained on datasets that are insufficiently represented in certain categories. On the other hand, some other classes have surplus samples due to the ready availability of data from these categories. Conventional solutions suggest undersampling of the majority class and/or oversampling of the minority class for balancing the class distribution prior to the learning phase. Though this problem of uneven class distribution is, by and large, ignored by researchers focusing on the learning technology, a need has now arisen for incorporating balance correction and data pruning procedures within the learning process itself. This paper surveys a plethora of conventional and recent techniques that address this issue through intelligent representations of samples from the majority and minority classes, that are given as input to the learning module. The application of nature-inspired evolutionary algorithms to intelligent sampling is examined, and so are hybrid sampling strategies that select and retain the difficult-to-learn samples and discard the easy-to-learn samples. The findings by various researchers are summarized to a logical end, and various possibilities and challenges for future directions in research are outlined.

KEYWORDS

class-imbalance problem, hybrid sampling, imbalanced data, oversampling, sampling, undersampling

1 | INTRODUCTION

Learning from imbalanced datasets results in a bias toward the majority class whose labeled samples are available in plenty as compared to the insufficiently represented minority class.¹ In data mining, factors that bring down the classifier performance are the intrinsic characteristics of the data and an uneven class distribution.² Lack of adequate data in the minority class results in a fuzzy and ever-varying decision boundary, leading to erroneous results. The problem is

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aggravated by the presence of noisy samples that are difficult to distinguish in a depleted minority population. The issue was first brought into limelight by Japkowicz³ as a significant research problem on its own. All real-world datasets have some degree of imbalance and a perfectly balanced dataset is difficult to find. Where the problem is of severe nature, the accuracy is generally high since almost all samples get classified as the majority class, and the poor performance of the depleted minority class does not affect the scores much. However, the matter assumes importance in circumstances such as when the minority class pertains to detection of rare cancer cells and a wrong diagnosis may prove to be fatal. Other examples where the minority population needs attention are software defect detection, fabric defect detection, and recognition of faces of non-celebrities vs celebrities.

The class-imbalance issue can be deduced from the classical probability statistics model presented by Bayes theorem, given by.

$$P(C_k|x) = \frac{P(x|C_k)P(C_k)}{P(x)} \quad (1)$$

The above equation that predicts the class of a sample x , can be interpreted as.

$$\text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{normalization} - \text{factor}} \quad (2)$$

The likelihood $P(x|C_k)$ and the prior $P(C_k)$ decide the value of the posterior probability that in turn decides the class of x . The largest posterior probability $P(C_k|x)$ that satisfies the condition $P(C_k|x) > P(C_i|x), \forall i \neq k$ indicates that the pattern x belongs to class C_k . In case of skewed class distributions, uneven class priors pre-exist even before the data are applied, and $P(C_{\text{Majority}}) \gg P(C_{\text{Minority}})$. The numerator will be very high in case of the majority class, leading to a biased decision since the posterior class probability will be automatically high irrespective of the outcome of the likelihood. The posterior class probability will be very low in case of the minority class even if the likelihood function is high. To avoid faulty decisions, most statisticians in history prefer to work with data having similar class priors.⁴

One reason for the surging interest in the topic of imbalanced datasets, in current times, is the fact that real-world datasets today are mostly crawled from the web that has variable sources of information.⁵ This causes an imbalance in the number of samples accumulated in different categories, unlike in the past when datasets were aggregated systematically by a human annotator. A comprehensive list of application domains where class-imbalance persists, such as medical diagnosis and network intrusion detection, is given in Reference 6. The number of samples per category is also subject to the commonality of occurrence of the category as observed for the cancer dataset where the number of “Benign” cases is almost double that of “Malignant” cases.⁷ Another reason that the issue is of current concern to the research community is that the number of categories has substantially spiked as compared to the last decade, primarily due to the advent of deep neural networks that can be trained on thousands of image classes using high performance computing machines.⁸ In cases where the Imbalance Ratio (IR) (ratio of the number of majority samples to the number of minority samples) is very high, all the samples of the minority may get classified as majority samples. Transforming the input data space by distance metric learning constrains class populations to restricted subspaces that improves classification performance, as observed by the authors in a recent work.⁹ Majority voting within an elite group of k -nearest neighbors of the transformed test sample decided its class label. However, distance metric learning becomes computationally complex when the number of classes is high as observed for the imbalanced Labeled Faces in the Wild (LFW) dataset in Reference 10. Here, the distance metric is learnt exclusively from sampled subspaces of the majority classes prior to the input space transformation. Also, transformations of the input space tend to improvise results only for distance-based classifiers and their variants; for the general group of classifiers, the improvement would not be significant since there is no change in the class distribution which is still uneven.

Balancing the class-distribution is thus essential for efficient learning and unbiased decision-making. As per the review in Reference 11, class imbalance can be resolved by one of several approaches namely (a) data-level techniques that involve sampling the dataset prior to learning, (b) algorithmic techniques that involve modifying the machine learning algorithm, and (c) cost-sensitive learning. The review concluded that while sampling achieved the best results in most of the cases, a combination of sampling and algorithmic techniques such as boosting is expected to enhance the classification performance. Our survey paper reviews state-of-the-art resampling strategies like Synthetic Minority Over-sampling (SMOTE) as well as hybrid algorithmic methods that incorporate resampling like SMOTEBoost which will be discussed in detail. Cost-sensitive learning assigns a higher weight to the misclassification cost pertaining to the minority class as compared to that of the majority class.¹¹⁴ In Reference 12, a weight of 1.0 was assigned to the minority class

and 0.1 to the majority class in order to correct a 1:9 class-imbalance. Susan et al¹³ proposed a class-weight equal to the ratio of total population to class population, for detecting lung nodule images. For the majority class (non-nodule images), this ratio was observed to be 1:1, while for the minority class (nodule images), the ratio was 6:1, thereby attributing a higher weight to the minority loss function. Zhou and Liu¹⁴ implemented oversampling, undersampling, and threshold moving with hard and soft voting schemes for cost-sensitive learning using neural networks. Their experiments concluded that threshold moving with soft ensemble was the best combination for tackling class-imbalance. Provided the misclassification costs are known beforehand, cost-sensitive learning is considered a better option as compared to sampling due to the obvious disadvantages related to loss of information while undersampling and the likely overfitting of data due to oversampling. These disadvantages can be minimized by employing intelligent sampling techniques as we discuss in this paper. However, it is difficult to know the misclassification costs beforehand, and not all machine learning algorithms have cost-sensitive implementations available. These issues together with the current availability of real-time data in huge volume, velocity and variety render sampling a practical choice over cost-sensitive learning.

Sampling of a dataset can be achieved in two ways: either by increasing the number of samples in the minority class to match the majority population (oversampling) or by cutting down the majority population to match the minority population (undersampling). Pruning the dataset by resampling is a natural choice for achieving the balance. However, at the same time, inter-class and intra-class diversity in the training data have to be maintained. This throws up a challenge in regard of the necessity of careful and intelligent pruning of imbalanced datasets such that the representation from all classes is balanced, and at the same time, inter-class and intra-class diversity is preserved. A careful selection of a subset of samples that carry class-discriminatory information would create a balanced distribution that achieves a better learning performance. Figure 1A illustrates the class-imbalance problem for a synthetic dataset. Some simplistic baseline resampling strategies for correcting the class-imbalance are also introduced; they will be perused in greater details in subsequent sections. The methods demonstrated include oversampling the minority class by SMOTE (Figure 1B), random undersampling (RUS) of the majority class (Figure 1C), shortlisting the borderline instances of both classes (Figure 1D), borderline instances with noisy samples removed (Figure 1E). The borderline instances are those that lie near the decision boundary and hence influence the decision-making. We will be exploring methods in our study where, once the borderline instances are obtained, they are oversampled to increase the final population. Most of the methods primarily differ on the basis of the “intelligence” of their sampling algorithm. The term “intelligent sampling” refers to the ability of the

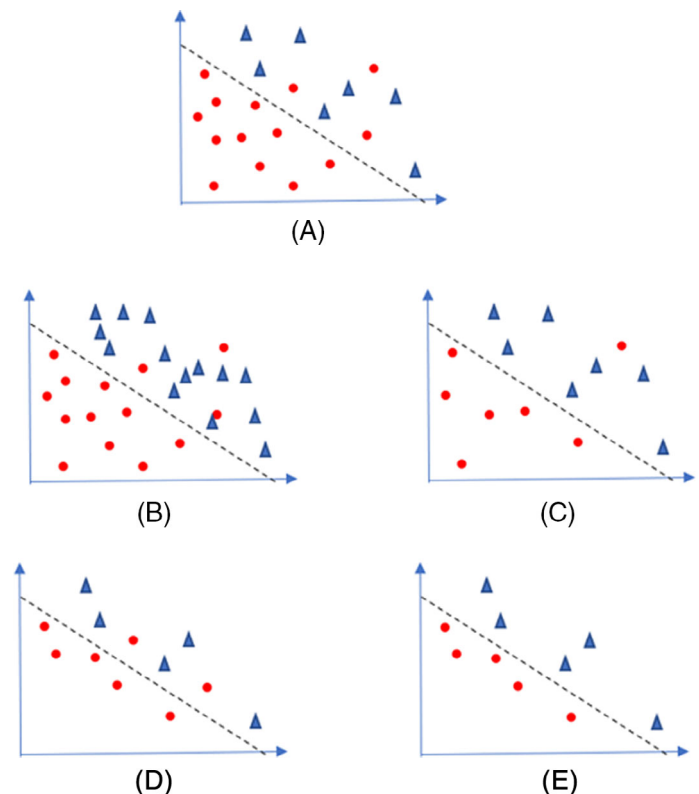


FIGURE 1 Demonstration of class-imbalance and some simplistic sampling strategies [Red Discs: Majority class, Blue Triads: Minority class]: (A) Original class-distribution (B) oversampling the minority class by SMOTE (C) random undersampling of the majority class (D) shortlisting the borderline instances of both classes (E) borderline instances with noisy samples removed

sampling algorithm to retain samples or create new samples that enhance class discriminatory information in order to induce a better classification performance. Interestingly, the number of samples in the final balanced population varies as per sampling strategy with a larger population in cases of oversampling (Figure 1B) and a depleted population in cases of undersampling (Figure 1C-E). However, as we shall see, the size of the final population has no bearing on the results. The performance depends on the class discriminatory information contained in the pruned population.

This paper discusses several approaches of pruning imbalanced datasets by resampling the majority and/or minority classes. The resampling strategies are roughly divided into undersampling and oversampling techniques and a set of hybrid techniques that employ both undersampling and oversampling. The term “intelligent sampling” is paid a lot of attention to, with an effort to understand the logic behind each method. The organization of this paper is as follows. Section 2 reviews various undersampling techniques. Section 3 discusses the vast diversity of oversampling techniques available in literature. Section 4 describes the hybrid sampling strategies. Section 5 discusses the implementation details and experimental setup for sampling algorithms that include a discussion on software tools, popular datasets and performance metrics. Section 6 summarizes the challenges and outlines the future directions of research and applications for imbalanced learning. Section 7 draws the final conclusions from our study.

2 | UNDERSAMPLING THE MAJORITY CLASS

Undersampling the majority class is a natural choice to begin with for solving the imbalanced class problem. The criteria for deciding which samples of the majority class should be deleted and which should be retained is what defines the undersampling strategy. The balancing achieved by simply cutting down the majority population is expected to eliminate any class-induced bias while learning, however at the cost of loss of information.¹¹³ How far this loss will affect the performance is a matter of study. Here, we review some methods that followed this line of research. The concept of cleaning the dataset by identifying and deleting noisy samples is an old concept. One of the significant works in this regard is that of Tomek,¹⁵ who defined Tomek links as a pair of samples belonging to two different classes whose distance is minimum. This could be interpreted in two ways: either both are borderline examples or one of the samples is noise. Hence the solution would be either to delete both the samples or delete the sample that belongs to the majority class. Several modern works in literature make use of Tomek links to balance their population as observed from Table 1 that outlines some distinctive sampling strategies.

In RUS demonstrated in Figure 1C, a small population of majority samples are selected randomly to match the minority population. Since no intelligent technique or reasoning was applied for the selection/rejection of majority samples, the loss of information in the form of useful majority samples that were not included, is likely to be higher.³ Clustering is another tool for cutting down the majority population. In Reference 32, the number of clusters of the majority class is set equal to the minority population so that a balance ensues. The cluster centers form the new majority population. Alternatively, substituting the cluster centers with the nearest neighbors is also considered, in the same work that proved to be a more viable approach. Mani and Zhang³³ conducted an investigation on the effect of class-imbalance on the k-NN classifier. They proposed NearMiss-1, NearMiss-2, and NearMiss-3 criteria for selecting majority class samples based on the average distance to the minority class samples; out of all three methods, NearMiss-2 performed best. NearMiss-2 selected only those majority samples that were close to the three farthest minority samples.

Often intelligent learning techniques are used in conjunction with, or as alternatives to, random sampling, for higher performance. Sometimes, undersampling is integrated into the learning process itself, as observed in the case of Reference 34 where two different approaches are presented. These are *EasyEnsemble* (learns a separate classifier for different subsets of samples from the majority class and combines all outputs using ensemble technique) and *BalanceCascade* (separate classifiers are trained sequentially on different subsets of samples from the majority class after removal of the correctly classified examples at each step). In Reference 35, the minority class is trained on a greater number of classifiers as an alternative to increasing the number of minority instances. The proposed Biased Random Forest introduces oversampling at the algorithmic-level instead of at the data-level. Some research works focus on undersampling while simultaneously transforming the data space. One such work is that of Dumpala et al.,³⁶ in which two samples are combined prior to training by simple concatenation and likewise for their respective class-labels that assume four different forms of (majority-majority), (majority-minority), (minority-majority) and (minority-minority). To cut-down on the contribution of majority samples, the combination (majority-majority) is reduced to (majority-majority_N), where majority_N constitutes a subset of the samples selected randomly from the majority class, the exact number of which is set equal to the number of minority samples (*N*). This data representation was termed the majority-constrained s2s data representation.

TABLE 1 Summary of some distinctive sampling strategies in literature

Method	Sampling strategy	Distinctiveness factor	Procedure of pruning the training set	Experimental details
SMOTE-TOMEK ¹⁶	Oversampling of the minority class followed by data cleaning	Detection of TOMEK links (majority-minority noisy pair of samples)	SMOTE followed by removal of TOMEK links to achieve data cleaning in both classes	- Domain: Data mining - Datasets: 15 UCI datasets - Classifier: Euclidean distance (Quantitative attributes) and Value Difference Metric (Qualitative attributes), Decision trees - Baseline approaches compared: SMOTE, SMOTE-ENN, RUS, CNN, Tomek, OSS, NCL
SMOTE-ENN ¹⁶	Oversampling of the minority class followed by in-depth data cleaning	Edited Nearest Neighbor Rule (ENN): Checking the class labels of three nearest neighbors of each majority sample	SMOTE followed by removal of the majority sample whose label is different from that of its three nearest neighbors	- Domain: Data mining - Datasets: 15 UCI datasets - Classifier: Euclidean distance (Quantitative attributes) and Value Difference Metric (Qualitative attributes), Decision trees - Baseline approaches compared: SMOTE, SMOTE-ENN, RUS, CNN, Tomek, OSS, NCL
One sided selection (OSS) ¹⁷	Undersampling the majority class by data cleaning	Retaining all the minority samples but removing selectively the noisy majority samples followed by removal of redundant negative samples (one-sided selection)	Construct TOMEK links and removing the associated noisy majority examples; remove redundant majority samples by removing the instances that are similar to the selected majority samples	- Domain: Data mining - Datasets: two data files related to oil spill detection project and five UCI datasets - Classifier: 1-Nearest Neighbor, Decision trees - Baseline approaches compared: using all training samples vs data reduction by removing redundant negative samples
Cluster-SMOTE ¹⁸	Oversampling the minority class after clustering	Independent cluster-wise oversampling	Form sub-clusters of minority class by k-means clustering algorithm and oversample each sub-cluster using SMOTE	- Domain: Intrusion detection - Datasets: Intrusion datasets collected from network traffic - Classifier: Rule based classification - Baseline approaches compared: SMOTE, RIPPER rule learning for intrusive traffic + Undersampling

(Continues)

TABLE 1 (Continued)

Method	Sampling strategy	Distinctiveness factor	Procedure of pruning the training set	Experimental details
Undersampling based on clustering (SBC) ¹⁹	Undersampling the minority class after clustering	Independent cluster-wise undersampling	Form sub-clusters of minority class by k-means clustering algorithm and undersample each sub-cluster based on value of imbalance ratio of the specific sub-cluster	- Domain: Data mining - Datasets: Artificial and real-world datasets (two UCI datasets) - Classifier: Neural network - Baseline approaches compared: RUS, NearMiss-2
Cluster-based oversampling by inflating small disjuncts (CBOS) ¹²	Oversampling clusters of both the minority and majority classes	Clustering of majority and minority classes separately followed by oversampling each cluster to address the within-class-imbalance along with the pre-existing inter-class-imbalance	Oversample the clusters of majority (to match the largest class) and then increase the minority class population to match the oversampled majority cluster population	- Domain: Data mining - Datasets: Artificial and real-world datasets (two UCI datasets) - Classifier: Decision trees and neural network - Baseline approaches compared: Oversampling, Undersampling, Cost modification
Safe-level-SMOTE ²⁰	Selective oversampling of the minority class	Compute a threshold called safe-level for every minority sample based on the number of minority instances in the set of its k-nearest neighbors	Generate synthetic minority samples “in the safe zone” nearer to a minority instance with larger safe-level	- Domain: Data mining - Datasets: two UCI datasets - Classifier: Decision trees, SVM, Naïve Bayes - Baseline approaches compared: SMOTE, Borderline SMOTE
Similarity-based undersampling and normal distribution based oversampling (SUNDO) ²¹	Independent oversampling of the minority class and undersampling of the majority class	Compute distance of candidate synthetic minority samples from the majority class (for oversampling); Compute similarity between pairs of majority samples to detect redundancy (for undersampling)	Consider the synthetic minority samples that have the largest distance from the majority class (oversampling); Eliminate one of the pair of similar majority samples (undersampling)	- Domain: Data Mining - Datasets: Artificial dataset and two industrial datasets - Classifier: SVM, Bayesian classifier, Decision trees, Self organizing map - Baseline approaches compared: SMOTE, informed SMOTE, informed clustering based oversampling
Distinct-Borderline (DBI) ²²	Oversampling of both classes followed by undersampling of the majority class	Create a separate boundary subset of majority and minority samples; Generate synthetic samples for both the minority class and the majority class in this boundary subset	Remove certain majority samples from the boundary subset that are similar to the minority samples in the subset	- Domain: Data mining - Datasets: six UCI datasets - Classifier: Decision trees - Baseline approaches compared: Random oversampling, SMOTE, Borderline SMOTE

(Continues)

TABLE 1 (Continued)

Method	Sampling strategy	Distinctiveness factor	Procedure of pruning the training set	Experimental details
Probability density function (PDF) estimation based Oversampling (PDFOS) ²³	Oversampling of the minority class based on estimated statistical distribution	PDF of the minority class estimated by the Parzen-kernel window function.	For every minority sample generate synthetic samples using the PDF assuming new samples share the same statistical properties	• Domain: Data mining • Datasets: six UCI datasets • Classifier: Radial Basis Function • Baseline approaches compared: Orthogonal Forward Selection (OFS), k-means
STA-SMOTE ²⁴	Oversampling of the minority class	Synthetic samples are generated by SMOTE whose adjustable parameters (N, k) are optimized by the State Transition Algorithm (STA)	Number of synthetic samples are restricted between the limits of minority population and majority population	• Domain: Data mining with industrial application • Datasets: four UCI medical datasets and aluminum electrolysis dataset collected from an aluminum industry in China • Classifier: Decision trees and neural network • Baseline approaches compared: PSO-SMOTE
Evolutionary Undersampling (EUS) ²⁵	Undersampling of the majority class by prototype selection	Removal of instances from the majority class based on the survival of the fittest chromosome	Fitness function is set equal to accuracy/Area Under Curve (AUC)/Geometric Mean (GM) of True Positive Rate and False Positive Rate (<i>AUC, GM gave equally good performance</i>)	• Domain: Data mining • Datasets: 28 UCI datasets (results averaged across all datasets) • Classifier: 1-Nearest Neighbor classifier • Baseline approaches compared: CNN, NCL, OSS, RUS, Tomek Links,
NRSBoundary-SMOTE ²⁶	Oversampling of the minority class based on the Neighborhood Rough Set (NRS) model with the aim that majority class performance is unaffected	Construct a BORDERLINE set comprising of those minority instances that are closer to the decision boundary and are in a rational neighborhood that do not affect majority samples	Synthetic samples generated based on the minority instances in the BORDERLINE set	• Domain: Data mining • Datasets: 15 UCI datasets • Classifier: C4.5 Decision trees, CART, k-NN, SVM • Baseline approaches compared: SMOTE, Borderline SMOTE, SMOTE-RSB
Under Sampling Strategy (USS) ²⁷	Undersampling of the minority and majority classes	Removal of noisy samples defined as misclassified instances whose nearest neighbors belong to the opposite class	Iterative removal of the weak instances and use of a correlation-based subset filter for obtaining the optimal sample subset	• Domain: Software engineering • Datasets: four software defect analysis datasets • Classifier: Decision trees • Baseline approaches compared: C4.5, Classification and Regression Trees (CART), reduced error pruning tree

(Continues)

TABLE 1 (Continued)

Method	Sampling strategy	Distinctiveness factor	Procedure of pruning the training set	Experimental details
SMOTE-Cosine ²⁸	Oversampling of the minority class	Compute the cosine similarity between each minority sample and all other minority samples to determine nearest neighbors	Sort the nearest neighbors of minority samples as per sum of ranks assigned by cosine similarity and Euclidean distance and generate synthetic samples for samples at the top	• Domain: Data mining • Datasets: 18 UCI datasets • Classifier: SVM with linear kernel • Baseline approaches compared: SMOTE
Adaptive semi-supervised weighted oversampling (A-SUWO) ²⁹	Oversampling of the minority class	Cluster minority samples by hierarchical clustering and oversample each sub-cluster considering its Euclidean distance from the majority class	Generate synthetic samples for the borderline cases in each sub-cluster	• Domain: Data mining • Datasets: 16 UCI datasets • Classifier: SVM • Baseline approaches compared: Random oversampling, SMOTE, Borderline SMOTE, Safe-level SMOTE, Cluster-SMOTE, MWMOTE, CBOS, SBC
Hybrid sampling with data space transformation ³⁰	Undersampling of the majority class followed by oversampling of the minority class followed by undersampling of the oversampled minority class	Sampling is followed by data space transformation of the pruned and balanced training dataset	Classifier is trained on the pruned and transformed dataset. The transformation matrix learned from the pruned training dataset is applied to the test sample for effective distance-based classification	• Domain: Data mining • Datasets: 12 UCI datasets • Classifier: 1-Nearest Neighbor classifier • Baseline approaches compared: ADASYN, MWMOTE, Borderline SMOTE, SSO_{Maj}-SMOTE-SSO_{Min}
CDSMOTE ³¹	Undersampling the majority class by class decomposition (CD) and oversampling the minority class by SMOTE	Reducing the dominance of majority class without information loss	Majority samples are clustered based on similarities by Class Decomposition and the minority class is oversampled by SMOTE to balance the distribution	• Domain: Data mining • Datasets: 60 datasets from KEEL repository • Classifier: SVM with linear kernel Baseline approaches compared: ADASYN, SMOTE, Class decomposition

The testing phase comprised of combining the test sample with the reference samples followed by voting for predicting the class label.

The disadvantage of RUS of the majority class is the possibility of loss of useful samples that could have helped discriminate between the majority and minority classes. Neighborhood Cleaning Rule (NCL) was proposed by Laurikkala³⁷ for data reduction of difficult datasets using Wilson's Edited Nearest Neighbor (ENN) distance rule.³⁸ The ENN rule states that those majority class samples whose two out of three nearest neighbors have different class labels are deleted. Apart from NCL, all undersampling methods tagged with (-ENN) use the above stated Wilson's ENN rule. NCL modifies Wilson's rule by applying it on the minority class as well. If the minority class samples have two out of three nearest neighbors differing in class labels, and if these nearest neighbors belong to the majority class, then these majority samples are also deleted. The research in NCL promotes ideas leading to modification of pre-existing distance and similarity rules for matching of templates by redefining these algorithms to behave differently for the minority and the majority classes. Recently, a number of sampling algorithms have emerged that are based on evolutionary algorithms for refining their population. Some of the undersampling techniques that follow this approach are discussed next. To tap the "usefulness" of samples toward class-discrimination, a cross-validation procedure was used in Sample Subset Optimization (SSO) proposed by Yang et al.³⁹ Optimized subsets of samples that minimize the k-fold cross-validation error, are short-listed by an evolutionary algorithm (both Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) are tested for the purpose). PSO⁴⁰ is a meta-heuristic algorithm that simulates bird flocking behavior to achieve an objective such as finding food. GA proposed by Goldberg⁴¹ is based on the principle of natural selection. Here, a population of chromosomes representing the unknown variables are refined, with the fittest individuals in each generation being selected for reproduction, cross-over and mutation, to generate the next generation of offspring. In the undersampling scenario, the objective is to find the optimal subset of training samples in the majority class that offers the maximum discriminatory information. The majority samples are then ranked based on the number of times they have been included in such optimized subsets and the most frequent of them all are selected as the representative samples from the majority class that balances the minority population. Another approach that used GA for instance selection was observed in Reference 42 where a cluster-based undersampling overcame the class-imbalance problem in bankruptcy prediction. Ant Colony Optimization was applied in Reference 43 for shortlisting significant majority samples of the DNA microarray data that suffers from severe imbalance problem. Randomly, the training set was divided into sample subsets. A statistical analysis of the results highlighted the majority samples that got selected with higher frequency. These high frequency samples were then combined with minority samples for learning. PSO was investigated in Reference 44 for example-based learning under the axiom that incorporating multiple good examples while learning leads to fast and good solutions. Emerging works such as⁴⁵ explore more obscure swarm intelligence techniques such as Stochastic Diffusion Search and Dispersive Flies Optimization for intelligent sampling of the majority population. It was observed here that the agents' activity is proportional to data redundancy in the dataset. Figure 2 shows the results of intelligent

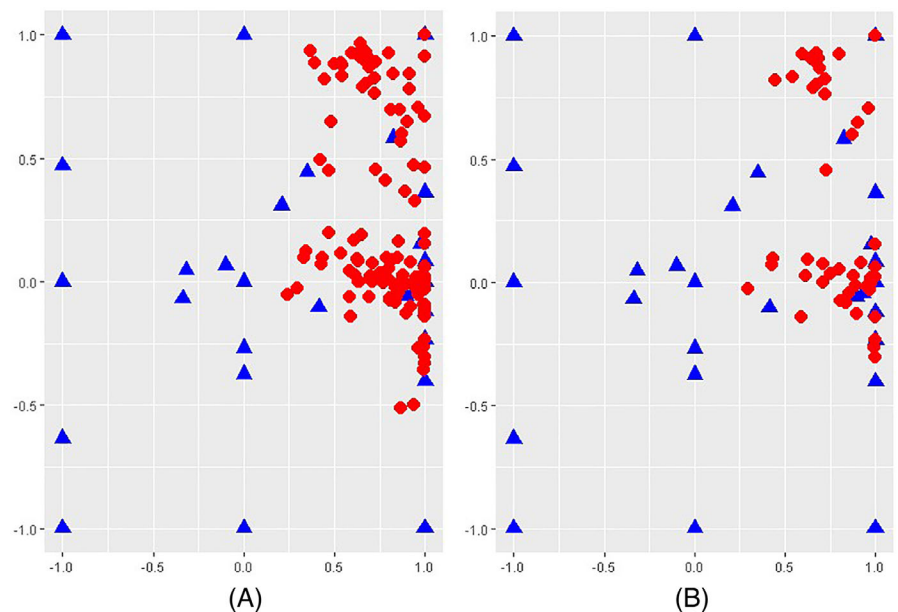


FIGURE 2 Demonstration of an intelligent undersampling scheme [Red Discs: Majority class, Blue Triads: Minority class]: (A) Original distribution of Ionosphere (B) Balanced distribution after intelligent undersampling of majority class by SSO (with Particle Swarm Optimization as the intelligent agent in the search space)

undersampling of the majority class by SSO-PSO, on the dataset *Ionosphere*. Majority and minority samples after pruning are confined to local regions with a few odd exceptions. The distribution of the pruned and balanced population in the case of SSO-PSO (Figure 2B) is intuitively observed to be more helpful for learning class boundaries than is possible with RUS, where random samples, irrespective of their distance from the minority samples, are deleted in order to balance the population.

RUS of the majority class involves simple programming that is executed in a fraction of seconds in a computer of configuration Pentium- 266 MHz 96 MB-RAM Desktop-PC, or higher. The undersampling algorithms that involve evolutionary algorithms like GA, PSO, and so on, for selecting the optimal subset, take up more time for execution depending on the size and dimension of the search space. In fact, the primary advantage of using RUS is that it is simple and fast. RUSBoost⁴⁶ combines RUS with AdaBoost to provide a simpler and faster version of its predecessor SMOTEBoost that combines the oversampling technique SMOTE with the AdaBoost classifier. While SMOTE involves some computational complexity in the form of interpolation to compute new synthetic samples, RUS involves simply deleting samples at random which is a much faster and simpler approach. SMOTE and SMOTEBoost are discussed in more details in the next section that is devoted to oversampling strategies.

3 | OVERSAMPLING THE MINORITY CLASS

Oversampling the minority class is another natural choice to solving the imbalanced class problem, by creating more instances of the underrepresented minority class. Random oversampling selects the minority samples to be replicated in a randomized manner. Several researchers prefer increasing the number of samples as opposed to removing samples via undersampling since the latter may lead to loss of some amount of information. The instances can be generated by either duplicating the minority samples or by introducing rotation and skew to perturb the data.⁴⁷ This approach however generally suffers from the side-effects of creation of random noise and unnecessary duplications if oversampling is not handled intelligently, and generally may lead to overfitting. One of the most popular works that emerged over time as a standard baseline is that of Chawla et al⁴⁸ who proposed the SMOTE Technique. It worked on the simple principle of creating new minority instances by interpolating neighboring minority instances. The aim was to introduce variation in the dataset. The interpolation process is demonstrated below for two minority instances p and q that are interpolated to create a new synthetic instance r as.

$$r = p + (q - p) \times \lambda \quad (3)$$

where λ is a random number in the range $[0,1]$. There are two adjustable parameters associated with SMOTE: the oversampling rate and the number of nearest neighbors. The advantages of SMOTE include simplicity of execution and robustness when applied to a variety of data imbalance problems. Since then several advancements over SMOTE and its extensions have been proposed in the past 15 years which have been comprehensively summarized in the recent survey by Fernández et al.⁴⁹ SMOTE expands the decision space for the minority class, improving the performance, while preventing overfitting. However, at the same time, the decision space for the majority class is contracted and the probability of misclassification for majority samples may rise. Hence researchers who propose extensions or variations of SMOTE may well take note of the fact, for effective classification of both the minority and majority classes. Combinations of SMOTE with AdaBoost classifier (SMOTEBoost⁵⁰) has led to improved learning performance. Boosting is an iterative learning algorithm that in each iteration assigns higher weights to misclassified examples. Since the misclassified examples, in case of imbalanced data, are more likely to belong to the majority class, a skewed distribution exists among the misclassified examples as well. Hence SMOTE is incorporated into the Boosting procedure by which, in each iteration, minority examples that are misclassified are oversampled as per the SMOTE algorithm, a classifier is learnt, and then these examples are discarded; this process repeats for every iteration. DataBoost-IM⁵¹ and RAMO-Boost⁵² were developed along similar lines. While DataBoost-IM oversamples both majority and minority classes in each boosting iteration, RAMO-Boost advocates ranked minority class oversampling in which the minority samples are ranked in each boosting iteration according to an underlying probability distribution so that the decision boundary is adaptively shifted toward the more difficult-to-learn instances. In Enhanced Structure Preserving Oversampling⁵³ and Mahalanobis distance-based oversampling,⁵⁴ the structure of the original input dataset is preserved while generating synthetic samples. Structure could pertain to either the covariance distribution or the underlying probability distribution.

Some of the significant and well-known extensions of SMOTE are described next. Each aims to apply some form of intelligence interwoven into the sampling process. Intelligence term here pertains to being aware of locations of other minority samples and/or majority samples while oversampling in order to create a new population that will define the class boundaries in a better manner. Borderline SMOTE is one of the first attempts toward constructing intelligent oversampling algorithms.⁵⁵ The motive is selective resampling that involves replication of those minority instances that are difficult to classify and lie near the decision boundary, as observed from the toy example in Figure 1D. First of all, the borderline minority examples that lie along the decision boundary are identified. Then the k nearest neighbors of each borderline minority instance are determined and new minority instances are synthetically generated by interpolating along the line connecting each borderline minority sample and each of its nearest neighbors, thereby strengthening the borderline minority population. In Adaptive Synthetic sampling,⁵⁶ the number of synthetic instances were determined individually for each minority sample using a weighted distribution proportional to the sample's closeness with the majority class. More number of synthetic instances were generated for those minority examples that proved to be difficult to learn, due to their greater proximity to the majority samples. The number of synthetic minority samples generated can be thus adaptively controlled. Both Borderline SMOTE and ADASYN focus on resampling those minority samples that are close to the decision boundary. Other type of SMOTE variations boast of newer interpolating techniques such as—those that involve interpolating with majority instances as well,⁵⁵ creating synthetic samples closer to the minority instance than any of its k nearest neighbors,⁵⁷ clustering-based techniques that consider the cluster centroids to be the new

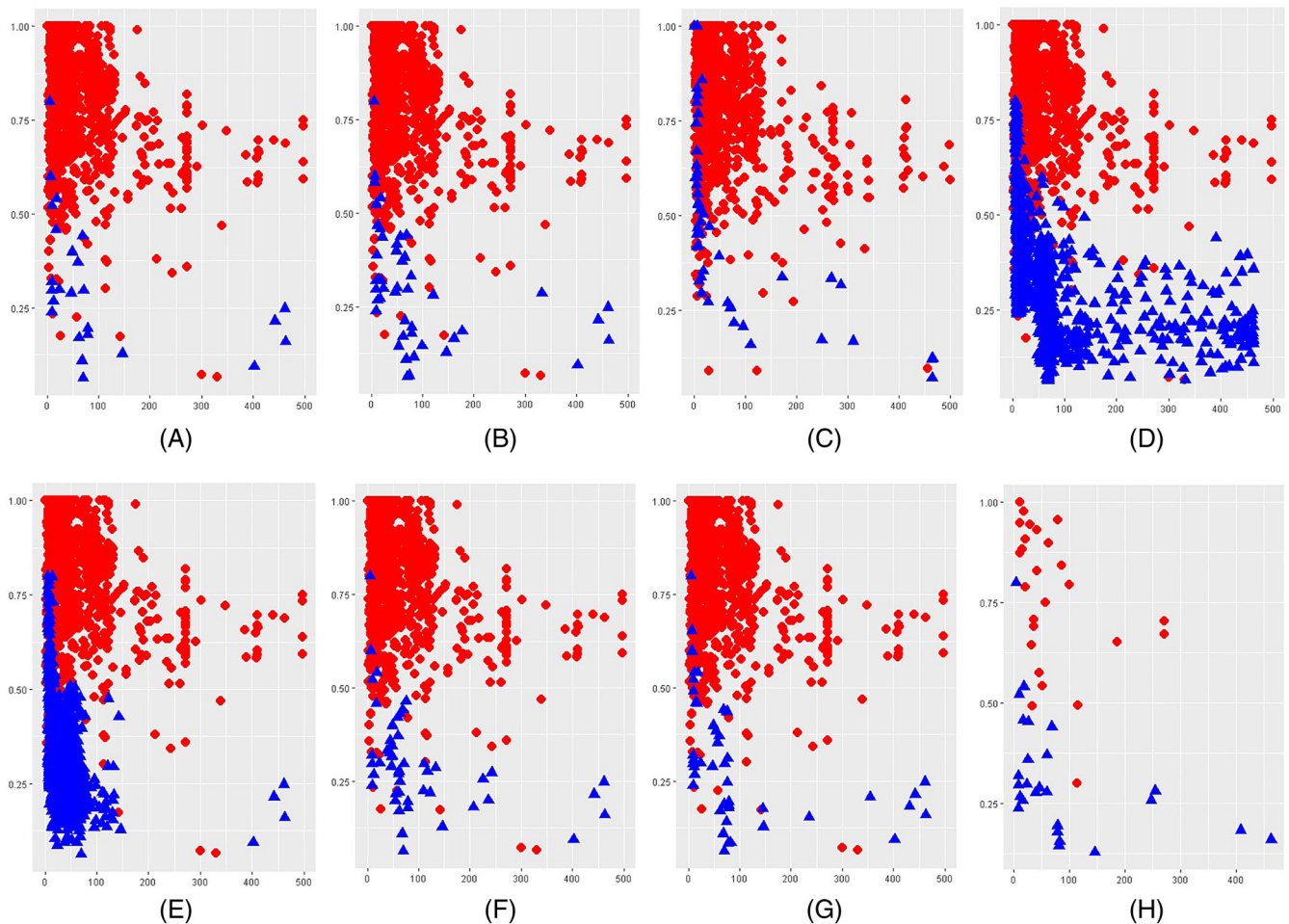


FIGURE 3 Balanced distribution achieved after applying some of the resampling methods (based on SMOTE oversampling and its variants) on the imbalanced *Page-blocks* dataset [Red Discs: Majority class, Blue Triads: Minority class]: in the order from Left to Right, (A) Original distribution (B) Distribution after SMOTE (C) Distribution after Borderline SMOTE (D) Distribution after ADASYN (E) Distribution after MWMOTE (F) Distribution after MAHAKIL (G) Distribution after SMOTE-RSB (H) Distribution after hybrid SSO_{Maj} -SMOTE- SSO_{Min} hybrid sampling strategy

synthetic minority instances⁵⁸ and estimation of probability distribution functions of the data.²³ The type of research covered in SMOTE and its extensions are summarized as: selecting the right minority samples, adaptively controlling the number of created instances for each minority sample, and defining the computation involved in generating the new synthetic samples.

As an inspiration from SMOTE and its variations discussed above, a new generation of works have emerged that deserve mention due to their new perspectives on oversampling, and it is some of these works that we review next. The first of these works is Majority Weighted Minority Oversampling Technique (MWMOTE)⁵⁸ that identifies “difficult” minority samples that are hard to learn, weights them as per respective distances to the majority class, and then creates the new synthetic instances by hierarchical clustering of the weighted informative minority samples. The new instances are generated from a cluster by interpolating two random members of the cluster. Biologically inspired solutions to oversampling are also found in literature. MAHAKIL⁵⁹ is based on the chromosomal theory of inheritance and was proposed for software defect detection. The minority instances are divided into bins, and samples with the same sequence number are averaged to form a representative synthetic sample. Another method, inspired by SMOTE, is certainty guided minority oversampling proposed by Zhang et al⁶⁰ in which minority samples are carefully oversampled along the decision boundary by checking for each instance if it increases the certainty of the complete dataset. Certainty of a dataset is interpreted to be the cumulative posterior class probability associated with the instances in the dataset. Only the synthetic instances that create a positive change in the certainty are allowed to be included in the resampled dataset. The approach is distinct due to its treatment of the majority class samples that are considered while generating synthetic samples of the minority class. Minority Oversampling in Kernel Adaptive Subspaces was proposed in Reference 61 as an intelligent sampling strategy. The synthetic samples here, are generated from various subspaces that are trained by competitive learning, and hence bear characteristics of the original input data and at the same time show some variation. Figure 3 demonstrates several resampling strategies for a real-world dataset including popular oversampling schemes such as SMOTE, Borderline SMOTE, ADASYN, MWMOTE, and MAHAKIL. The complex algorithms of ADASYN and MWMOTE are observed to create a large population of minority instances, with well-defined class boundaries. The minority instances in the case of ADASYN (Figure 3D) are noted to be more scattered while the population is found concentrated in the case of MWMOTE (Figure 3E). The real test for an intelligent sampling strategy is when the trained classifier is presented with new data. The computational complexity of oversampling techniques depends on the complexity of the intelligent sampling incorporated. SMOTE and simple variants that involve interpolation of existing samples have fast execution time while those that involve careful selection of the minority samples to be replicated are more computationally intensive.

4 | HYBRID SAMPLING STRATEGIES

The minority samples are retained as it is, in most of the undersampling methods discussed in the previous section. These minority samples, however, could be noisy or borderline or redundant data that are seldom removed out of concern for decrementing an already diminished minority population. This section reviews some of the methods that advocate hybrid sampling strategies by combining oversampling and undersampling in their bid to jointly consider both the underrepresented minority class and the overrepresented majority class. Oversampling may sometimes lead to overfitting. Hence it is advisable to clean the data by undersampling prior to the oversampling. Most of these hybrid strategies use SMOTE as the oversampling technique. Hu et al⁶² implement the undersampling part by PSO. Their hybrid sampling strategy called SMOTE-PSO was applied successfully for identifying malicious web domains. A related work is SMOTE-SSO that sequentially performs oversampling of the minority class and undersampling of the majority class.⁶³ SMOTE-RSB combines SMOTE with an editing technique based on the Rough Set Theory.⁶⁴ The synthetic samples generated by SMOTE were evaluated for their similarity with the majority class samples. Those synthetic samples whose similarity index was high, were deleted from the training set since they do not contribute to class-discriminatory information. Hence this method qualifies as a hybrid strategy since it oversamples and then cleans the synthetically generated minority data by undersampling. SMOTE was integrated with data gravitation-based classification (DGC) in Reference 65, a physical-inspired classification model that fails under conditions of class imbalance; the hybrid was named SMOTE-DGC. The hybrid method SSO_{Maj} -SMOTE- SSO_{Min} ⁶⁶ involves careful selection of the population of both classes, and only representative samples that carry discriminatory information are retained on both sides. An evolutionary algorithm (PSO) is used for the selection. The three-step data pruning process in SSO_{Maj} -SMOTE- SSO_{Min} constitutes of: (1) undersampling the majority, (2) oversampling the minority, and (3) undersampling the oversampled minority. The results marked an improvement over the baseline undersampling and oversampling techniques. All the above methods that involve hybridization of

SMOTE with intelligent undersampling, facilitate careful selection of both the majority and minority population that is a better proposition than careful selection of the majority samples alone. Figure 3 shows the distribution of the resampled *Page-blocks* dataset after application of several sampling techniques. The red polygon shaped markers indicate the majority class while the blue triads indicate the minority class. The original distribution shown in Figure 3A highlights the class-imbalance problem. SMOTE and its variants are observed to augment the minority population though not at a huge scale as in ADASYN and MWMOTE that agrees with our discussion in Section 3. The results of the hybrid sampling strategies of SMOTE-RSB and SSO_{Maj} -SMOTE- SSO_{Min} are also shown. SMOTE-RSB shows a variation in the minority population while the majority population is intact since both oversampling and undersampling happens in the minority class only. All these methods depict some changes from the original distribution, even though the changes may be incremental. One observation is that not all methods implement the perfect balancing act. ADASYN, MWMOTE, and SSO_{Maj} -SMOTE- SSO_{Min} yield a balanced distribution with equal members in both classes. The final balanced population of SSO_{Maj} -SMOTE- SSO_{Min} is observed to be sparse as compared to all other methods, with the size of the final population dependent on the size of the minority population in the original distribution. The balanced datasets are used to train suitable classifiers, some popular choices being Support Vector Machine (SVM), logistic regression, naïve Bayes, C4.5 decision trees, and k-nearest neighbor classifiers.

In Reference 67, both undersampling and oversampling were performed, with the oversampling being achieved by fuzzy logic; and elitist selection, recombination and mutation operators facilitating the undersampling part. A hybrid undersampling-based ensemble approach HUSBoost⁶⁸ finds special mention here since it achieves balancing by simple RUS and combines several classifiers through a soft voting approach to decide on the class of the test label. It is interesting to note that the first step in the HUSBoost algorithm is data cleaning by removal of noisy data using Tomek Links. A recent work by Li et al⁶⁹ uses entropy as a measure for retaining the difficult-to-learn samples and discarding the easy-to-learn samples. The resulting entropy-based hybrid sampling encourages a new research pattern that involves statistical machine learning for finding discriminatory samples. A recent work that addresses multi-class imbalance is Evolutionary inversion of class-distribution in overlapping class regions proposed by Fernandes and de Carvalho.⁷⁰ This method focuses only on the minority instances lying in the overlapping regions between classes since these have more class-discriminatory information. A multi-objective evolutionary algorithm is used to determine the population for each individual classifier in an ensemble of classifiers with a greater selection of minority instances being from the overlapping region. This method ensures diversity learning since the classifiers have different views of the dataset. However, the computational complexity is high due to the iterative evolutionary process executed for each classifier in the ensemble. Considering time complexity, hybrid strategies have more computational overhead due to the sequential process flow integrating multiple sampling schema. Parallelizing the sampling strategies in hybrid approaches, is a future research direction. Individual attempts at parallelizing generic sampling can be observed in References 71,72.

Some distinctive approaches to sampling in literature, other than those described above and in the previous sections, are outlined in Table 1 along with the concept used in each approach that makes it distinctive. The implementation details are also summarized for each method in the last column of Table 1. This includes the domain, datasets, classifier, and baseline methods compared with, for the reference of the readers.

5 | EXPERIMENTAL SETUP FOR IMBALANCED LEARNING

5.1 | Software tools

Python contains several useful libraries such as *numpy*, *scipy*, and *scikit-learn* that are used by programmers implementing data mining algorithms. These libraries contain inbuilt functions that are helpful for classification and decision-making. Sampling algorithms were implemented in the past in C/C++/JAVA languages that involve coding from scratch. This offers the programmer more flexibility in executing a sampling technique rather than relying on what the pre-built functions have to offer. Some recent works have adopted the “code from scratch” methodology such as SSO -PSO and SSO_{Maj} -SMOTE- SSO_{Min} that are coded in JAVA. The R language is also popular among data mining enthusiasts and contains functionalities such as training set selection, discretization, feature selection, and so on, that aid in classifying imbalanced datasets.⁷³

Modern toolboxes specialized for tackling imbalanced datasets have been introduced that offer a limited range of flexibility of choice of parameters to the programmer. *Imbalanced-learn* is an example of an open-source Python toolbox dedicated for learning from imbalanced datasets.⁷⁴ State-of-the-art methods for oversampling, undersampling, and

hybrid sampling such as SMOTE and Tomek Links are included in this toolbox. An open-source JAVA-based software tool is *Knowledge Extraction based on Evolutionary Learning (KEEL)* that incorporates evolutionary algorithms for classifying imbalanced datasets.⁷⁵ *KEEL* incorporates 45 algorithms for imbalanced classification that include cost-sensitive learning, ensemble, and resampling. *Multi-imbalance*⁶⁰ is an open-source software for multi-class imbalanced datasets. This toolbox includes 18 algorithms for multi-class learning in an imbalanced scenario. Ensemble techniques such as AdaBoost, DECOC, and DOVO are adapted for the multi-class scenario by incorporating sample weighting and intelligent combination of base classifiers. A performance analysis by the authors proved that DECOC and DOVO are the best software tools in *Multi-imbalance* for classification of multi-class imbalanced datasets. DECOC or Diversified Error Correcting Output Codes was proposed by Bi and Zhang⁷⁶ and DOVO is the Diversified One-vs-One strategy proposed by Kang et al⁷⁷ for multi-class classification. DECOC is a multiple classifier or ensemble system. Codewords are constructed for multiple classes based on the largest distance between the classes and a dichotomy classifier translates the codes into class labels. DOVO, on the other hand, selects the strongest classifier for a subset of data as the one with the lowest error rate. The subset is formed by selecting samples from each pair of classes. The outputs of the strongest classifiers for all subsets are combined to make the final prediction.

5.2 | Evaluation metrics

The performance evaluation metrics considered significant in case of skewed class distributions are: Area Under the Receiver Operating Characteristics Curve (AUC), G-mean, and the F1-score derived from the precision-recall values computed from predicted class labels, since these are more informative on the scores of the minority class.⁷⁸

F1-score is computed from precision and recall values as

$$F1 - score = \frac{2(\text{precision} \times \text{recall})}{(\text{precision} + \text{recall})} \quad (4)$$

where, precision and recall values are computed from the number of True Positives (TP), False Positives (FP), False Negatives (FN) as

$$\text{precision} = \frac{TP}{TP + FP} \quad (5)$$

$$\text{recall} = \frac{TP}{TP + FN} \quad (6)$$

The G-mean index is also used by some researchers.²¹ This index is defined as a function of the True Positive Rate (TPR) and the True Negative Rate (TNR) that are defined as follows.

$$G - mean = \sqrt{TPR \times TNR} \quad (7)$$

$$TPR = \frac{TP}{TP + FN} \quad (8)$$

$$TNR = \frac{TN}{TN + FP} \quad (9)$$

A g-performance measure was recommended for evaluating classification performance of imbalanced datasets in Reference 17, which is defined as the geometric mean of the percentages of majority and minority samples that have been correctly classified. The values of percentage accuracy of correct classification are often misleading for imbalanced datasets, since the minority samples are few in number, and even when all the minority samples are misclassified as majority, the percentages may still be uncomfortably high due to the entire majority population being correctly classified. Following this cue, research in Reference 79 focuses on devising new metrics that measure the state of imbalance in a more interpretable manner. The new metrics are Individual Bayes Imbalance Impact Index (IBI³) at instance level, and Bayes Imbalance Impact Index (BI³) at a data level. These measures reflect the extent of imbalance from the perspective of the minority population with respect to the whole dataset. While IBI³ was found to increase consistently with increase in accuracy, BI³ was found consistent to variations in F1-score.

TABLE 2 Some popular datasets, their Imbalance Ratios (IR) and reported results

Dataset	Minority class	Majority class	Minority samples	Majority samples	Imbalance ratio (IR)	Method (reported F1-score/AUC, classifier)
<i>Abalone</i> ⁵⁸	Class “18”	Class “9”	42	688	16.38	MWMOTE (F1-score: 0.394, Neural Network) ADASYN (F1-score: 0.298, Neural Network) SMOTE (F1-score: 0.441, Neural Network) RAMOBoost (F1-score: 0.439, Neural network) MWMOTE (F1-score: 0.608, k-Nearest Neighbor) ADASYN (F1-score: 0.440, k-Nearest Neighbor) SMOTE (F1-score: 0.501, k-Nearest Neighbor) RAMOBoost (F1-score: 0.662, k-Nearest Neighbor) MWMOTE (F1-score: 0.357, Decision tree) ADASYN (F1-score: 0.360, Decision tree) SMOTE (F1-score: 0.306, Decision tree) RAMOBoost (F1-score: 0.281, Decision tree)
<i>Yeast</i> ²⁶	MIT	All other classes	244	1240	5.08	SMOTE (F1-score: 0.573, Decision tree) Borderline SMOTE (F1-score: 0.527, Decision tree) SMOTE-RSB (F1-score: 0.5577, Decision tree) NRSBoundary SMOTE (F1-score: 0.574, Decision tree)
<i>Haberman</i> ¹⁶	Class “Die”	Class “Survive”	81	221	2.72	RUS (AUC: 0.684, Decision tree) CNN (AUC: 0.583, Decision tree) TOMKEK (AUC: 0.696, Decision tree) OSS (AUC: 0.620, Decision tree) SMOTE (AUC: 0.754, Decision tree) SMOTE+TOMKEK (AUC: 0.784, Decision tree) SMOTE+ENN (AUC: 0.770, Decision tree)
<i>Pima Diabetes</i> ⁵⁸	Class “1”	Class “0”	268	500	1.865	MWMOTE (F1-score: 0.679, Neural Network) ADASYN (F1-score: 0.683, Neural Network) SMOTE (F1-score: 0.668, Neural Network) RAMOBoost (F1-score: 0.660, Neural network) MWMOTE (F1-score: 0.621, k-Nearest Neighbor) ADASYN (F1-score: 0.616, k-Nearest Neighbor) SMOTE (F1-score: 0.636, k-Nearest Neighbor) RAMOBoost (F1-score: 0.628, k-Nearest Neighbor) MWMOTE (F1-score: 0.600, Decision tree) ADASYN (F1-score: 0.561, Decision tree) SMOTE (F1-score: 0.582, Decision tree) RAMOBoost (F1-score: 0.578, Decision tree)
<i>Statlog (vehicle silhouette)</i> ⁵⁸	Class “Van”	All other classes	226	720	3.18	MWMOTE (F1-score: 0.920, Neural Network) ADASYN (F1-score: 0.857, Neural Network) SMOTE (F1-score: 0.917, Neural Network) RAMOBoost (F1-score: 0.916, Neural network) MWMOTE (F1-score: 0.867, k-Nearest Neighbor) ADASYN (F1-score: 0.856, k-Nearest Neighbor) SMOTE (F1-score: 0.864, k-Nearest Neighbor) RAMOBoost (F1-score: 0.854, k-Nearest Neighbor) MWMOTE (F1-score: 0.898, Decision tree) ADASYN (F1-score: 0.896, Decision tree) SMOTE (F1-score: 0.890, Decision tree) RAMOBoost (F1-score: 0.883, Decision tree)
<i>Statlog (vehicle silhouette)</i> ²⁹	Class “Van”	All other classes	226	720	3.18	Random oversampling (F1-score: 0.955, SVM) SMOTE (F1-score: 0.953, SVM) Borderline SMOTE (F1-score: 0.959, SVM) Safe-level SMOTE (F1-score: 0.954, SVM) Cluster-SMOTE (F1-score: 0.858, SVM) MWMOTE (F1-score: 0.948, SVM) CBOS (F1-score: 0.955, SVM) SBC (F1-score: 0.917, SVM) A-SUWO (F1-score: 0.961, SVM)

5.3 | Some popular datasets for imbalanced learning and reported results

Two popular repositories for smaller sized real-world datasets or toy datasets, as they are sometimes called, are the UCI repository⁸⁰ and the online platform Kaggle. Examples of smaller sized real-world datasets used by researchers popularly for implementing imbalanced learning algorithms include *Yeast* (Gao et al,²³ Hu and Li²⁶), García and Herrera²⁵), *Ionosphere* (Hu and Li²⁶), *Abalone* (Barua et al,⁵⁸ García and Herrera²⁵), *Pima Diabetes* (Jo and Japkowicz,¹² Barua et al,⁵⁸ Gao et al,²³ García and Herrera,²⁵ Hu and Li²⁶), *Wisconsin Breast Cancer* (Jo and Japkowicz,¹² Pengfei et al²²), *Haberman* (Batista et al,¹⁶ Bunkhumpornpat et al,²⁰ Gao et al,²³ García and Herrera,²⁵ Hu and Li²⁶), among a few. *Pima Diabetes* and *Haberman* are observed to be the most popular choices among most researchers. Kaggle contains Big Data examples as well such as “The Nature Conservancy Fisheries Monitoring”, “European Soccer database” and “Intel & MobileODT Cervical Cancer Screening” that are examples of imbalanced datasets with uneven class distributions. The *KEEL* data repository⁷⁵ contains imbalanced datasets with imbalance ratios ranging from 1.5 to 9, with each dataset partitioned using 5-fold stratified cross-validation.

Datasets with binary classes are preferred by researchers for the experiments on imbalance learning. The reason is that most learning algorithms are written from the perspective of the inter-relationship of majority and minority class samples and their distances from the decision boundary. A single decision boundary is convenient to consider, and multiple decision boundaries, as in the case of real-world datasets, only complicates the algorithm and exponentially increases the computational complexity. Multi-class datasets are thus converted by most researchers to the binary class form by either considering any two differently populated classes, or by grouping classes into two main categories namely, majority and minority, by relatively comparing the class populations. Imbalance ratio (IR) is the ratio of majority and minority class population and its value indicates the extent of the class-imbalance problem for different datasets.³⁰ The IR of some popular datasets used for imbalanced learning in data mining are shown in Table 2 along with reported results; these IR values range from 1.86 to 57.63. Two sets of results extracted from two independent works are shown for the *Statlog (vehicle silhouette)* dataset to highlight the variety in the choices of classifiers and baseline approaches. The F1-score and AUC performance measures are the most popular choices as evident from the readings in Table 2. Among all methods, RUS, SMOTE, MWMOTE, and Borderline SMOTE appear to be the most popular baseline approaches. The scores of SMOTE are observed to be consistently high. The decision tree classifier is observed to be the most popular choice of the classifier followed by SVM. The k-Nearest Neighbor and 1-Nearest Neighbor classifiers perform well in Euclidean space and are sometimes used by researchers to highlight the worst-case scenario.

However, a high IR may not necessarily mean poor performance of the minority class, since a lot depends on the decision boundary between the majority and minority classes; and a well-defined decision boundary yields high performance for both classes despite of the minority samples being present in small numbers.

6 | FUTURE DIRECTIONS, APPLICATIONS, AND CHALLENGES

This section surveys the current challenges faced by data mining researchers working with imbalanced datasets and identifies future directions of research in imbalanced learning. We especially draw attention to open-ended research problems and potential application areas that are yet to be explored from the perspective of imbalanced learning.

6.1 | Exploration of classifier models

The response of individual classifiers needs to be studied under the impact of skewed class-distributions. An investigative study is conducted in Reference 81 to determine the effect of class-imbalance on various classifier models. It was found that among all classifiers, SVM is found least affected by the imbalanced class distribution, and oversampling tends to worsen the performance for most classifiers. On the other hand, the C4.5 decision tree classifier is most affected by the imbalance problem and responds well to imbalance treatment strategies like sampling. The survey by Branco et al⁸² investigates the side-effects of imbalance conditions in classification and regression tasks in predictive analytics. The study suggests improvisation of the standard evaluation metrics to focus on the rare case examples while training since these would be more class-informative. Investigators could follow the cue of Zhang et al⁸³ where a thorough investigation on 30 real-world datasets using six classifiers was done to prove the superiority of their SMOTE extension over SMOTE. In

a recent work, Raghuwanshi and Shukla⁸⁴ applied SMOTE for oversampling the minority class with the class-specific Extreme Learning Machine (ELM) as the classifier. The overall product is presented as SMOTE-CSELM, a new classifier that handles class-imbalance using SMOTE and has class-specific regularization parameters. This inspires ideas for modification of existing machine learning algorithms to integrate class-specific imbalance treatment, and present it as a new generation of classifier models.

6.2 | Handling multi-class imbalance

Another important open research problem is that of multi-class imbalance.⁶⁰ Most of the solutions offered so far are for binary class datasets where the majority vs minority class boundary is neatly defined.⁸⁵ Experiments in Reference 14 proved that, in most cases, these solutions were ineffective for handling imbalance in multi-class datasets. In the multi-class case, a divide-and-conquer scheme is still the norm, solved by dividing the multi-class problem into several two-class problems and aggregating the result of the individual classifiers.⁶ The detection of rare events is also an unexplored area from the perspective of class-imbalance.⁸⁶ The performance of the sampling algorithm is heavily dependent on the definitions of class regions for multi-class datasets. Future works need to do away with the concept of binarized classes and need to work out a standard protocol for applying resampling strategies in a multi-class scenario without redefining the class boundaries.

6.3 | Open research problems and future application areas

An insightful survey solely dedicated to addressing open research issues and challenges for imbalanced datasets is.⁸⁷ The survey identifies Big Data analytics, social media, and computer vision as vital application areas. The open research issues yet to be addressed from the perspective of imbalanced learning, as per the authors, are stated to be regression, learning from non-stationary data streams and multi-class problems. The survey in Reference 88 draws attention to a number of factors that aggravate the class-imbalance problem such as the existence of small disjuncts within a class, small sample size, and class overlapping occurrences. The application of imbalanced learning algorithms covers a wide spectrum of research. Text classification is a research area that has very few works devoted to correcting the class-imbalance.⁸⁹ Class-imbalance occurs when the number of text samples from a particular category is significantly less as compared to the other categories. Often the problem is not addressed and most researchers ignore the class-imbalance issue occurring in text-based analysis. The problem was noted in a recent work on twitter-based sentiment analysis where varying distributions of positive, negative, and neutral tweets in twitter datasets cast an ambiguity on the applicability of different sentiment lexicons such as SentiWordNet, AFINN, and Vader for the fuzzy classification process.⁹⁰ A bidirectional sampling based on k-means clustering algorithm was proposed in Reference 91 to handle the class-imbalance in a text-based corpus. SMOTE was applied for oversampling the minority cluster obtained by k-means algorithm. Bioinformatics projects that involve RNA and DNA identification, protein localization, prediction, and profiling often tackle imbalanced datasets when comparing human data with that of other species.³⁹ The applicability of resampling strategies for the bioinformatics field still needs to be explored in its entirety.

6.4 | Challenges in computer vision

In computer vision, deep learning is the current trend for learning from huge image databases.⁹² The AlexNet pre-trained network is trained on the 1000-class ImageNet database.⁹³ Addressing the class-imbalance problems in large-scale computer vision problems needs immediate attention, considering the information boom in surveillance sector that is an inherent part of our society now. Imbalanced learning algorithms specific to the field of Computer Vision need to be devised since resampling strategies are not practical in situations when the computational complexity is high, the number of classes is too high and the class-distribution profile is extremely uneven, as is the situation with most real-world vision tasks. Data augmentation of the minority class was explored in Reference 94 as a simple but effective solution to correct class-imbalance in Benign vs Malignant cancer classification for the BREAKHIS cancer dataset. Data augmentation is equivalent to oversampling, in computer vision. The term pertains to the synthetic generation

of new images by affine transformation (scaling, rotation, horizontal/vertical flips and shifts, shear, and so on) of the original images in the dataset. In Reference 95, deep learning, data augmentation, and Bag-of-Visual Words representation are analyzed with respect to their ability to handle class-imbalance. It was noted that as compared to the other approaches, deep convolutional neural networks have inherent properties that naturally tackle class-imbalance in an effective manner. A similar observation was made in Reference 96 in which a new cost function was proposed to improve the response of deep learning networks to class-imbalance. The deterioration of classification performance due to the heavy class-imbalance in the LFW facial image dataset containing 1680 classes, is highlighted in Reference 97. While some celebrities have only one or two images, others have hundreds of photographs due to easy availability of images of these celebrities, leading to a biased decision curve. A very interesting study on the effect of class-imbalance on Convolutional Neural Networks in Reference 98 concludes that oversampling is the best solution for balancing the dataset and that application of undersampling depends on the degree of imbalance. The behavior of deep convolutional networks needs to be investigated in more depth for imbalanced learning, both for networks trained from scratch and the pre-trained networks that are trained on millions of images and fine-tuned on smaller (and imbalanced) datasets.

6.5 | Challenges in big data applications

Another field where class-imbalance problems spike to mammoth dimensions is that of Big Data comprising of data in huge Volume, Velocity, Variety, and Variability, especially the confluence of multimodal sensor IoT data in the backdrop of development of smart cities.⁹⁹ Most of the researches involving new sampling algorithms and balancing techniques focus on datasets prevalent in data mining such as those available from UCI repository,⁸⁰ some of which are listed in Table 2 and covered in Section 5.3. An immediate challenge to be addressed by researchers tackling imbalanced learning is the applicability of the existing algorithms on real-world data such as large image, text and video databases, and sensor-based Big Data applications for Internet of Things (IoT). The continuous stream of multimedia data from electronic surveillance systems needs to be processed in real-time. The volume, velocity, and variety offered by such huge datasets are a challenge to the conventional machine learning algorithms that are mostly trained on toy datasets limited to a few thousands of samples.¹⁰⁰ However, to handle large-scale data containing millions of samples, a framework like MapReduce is required to distribute the computing among smaller computing elements. A recent study in Reference 101 advocated the distributed procedure followed in the MapReduce framework to aid in tackling class-imbalance in Big Data as compared to the conventional solutions that seem to fail for Big Data applications. Distributed evolutionary undersampling is explored in Reference 102 for Big Data. In this scheme, evolutionary undersampling is performed on smaller subsets of the training data using MapReduce, and each undersampled subset is trained on a decision tree classifier. An ensemble approach aggregates the outputs of the individual classifiers. Experiments on Big Data were successfully conducted in Reference 103 for achieving resampling with MapReduce and classification using the Random Forest classifier. Oversampling, undersampling, and cost-sensitive learning were investigated, and the studies concluded that when the number of mappers is increased, the processing time becomes faster at the cost of a slight drop in accuracy. A GPU-based implementation of SMOTE achieved good results in Reference 104 that paves the way for large-scale computing machines for sampling Big Data. Information filtering is essential under such circumstances with a clear understanding of which samples are significant and are to be retained. Han et al¹⁰⁵ recently confirmed this theory by introducing a distribution-sensitive undersampling for IoT datasets that retained the majority samples near the decision boundary and deleted noisy and stable majority samples. The minority samples near the boundary were replicated for achieving oversampling.

6.6 | Imbalance in streaming data

One topic that needs to be researched more is the co-occurrence of class-imbalance with other problems encountered in the learning phase. A pioneering work in this regard is that of Wang et al¹⁰⁶ who investigated the co-occurrence of class-imbalance with concept drift in online learning. Concept drift is the problem of dynamically shifting probability distributions associated with non-stationary data streams. Application of the conventional remedies for concept drift may aggravate the prevailing class-imbalance. It would also be difficult to attribute the large classification error to either cause. A thorough experimental study in Reference 106 concluded that imbalance remedies such as oversampling enhances

performance but interferes with drift detection. Research on real-time learning from streaming data is still in its evolution stage. The issue is complicated since class definitions change dynamically over time as proved for the XML data streams in Reference 107. The circumstance (concept drift) is tackled by Brzezinski and Piernik through incrementally mined subtrees and a suitable similarity measure. One of the earlier works that tackles population drift and changing class dynamics is that of Kelly et al.¹⁰⁸ The paper identifies three types of population drifts: those associated with class priors, posterior probabilities, and class probabilities, out of which change in posterior class probabilities is most indicative. The classification rule is updated by including the effect of new data points and down-weighting the effect of the older points. An advisable approach to tackle real-time analytics is the online updation of the learning model as fresh data is streamed online, rather than training a new model from scratch every time a fresh batch of data is released.¹⁰⁹ In this spirit, a low-dimensional linear subspace was estimated in Reference 110 as a low-rank matrix completion problem that reconstructed the input signal as a byproduct. This method incorporates missing and noisy entries in the data stream. Skewed class distribution in data streams is handled by importance sampling in Reference 111. The majority class, in this case, is undersampled with replacement while minority instances are generated as per importance weights attached to specific feature groups. The group weighting scheme is dynamic in nature and evolves with time over shifting time windows across the data stream. Class-imbalance for online learning is explored in Reference 112 for the multi-class scenario. A weighted online sequential ELM with kernel was proposed for the purpose that learns either in a one-by-one and chunk-by-chunk manner. The future research in devising new sampling strategies would thus see the involvement of high computing machines with cluster computing for high volume multimodal data, and real-time management of streaming data with shifting class priors.

7 | CONCLUSIONS

Imbalance in class distribution in modern-day real-world datasets is a common and urgent problem affecting researchers working in all realms of pattern recognition and machine learning. The issue is mainly overlooked by most researchers, though undeniably the learning performance is known to surge higher if some form of balancing trick is applied for balancing the class-distribution prior to the learning phase. This paper reviews the state-of-the-art balancing strategies, specifically sampling that transforms the data space to create a balanced training distribution prior to the learning phase. The survey covers basic concepts, implementation details, and future challenges related to the state-of-the-art in resampling. Undersampling the majority class is a good balancing strategy from the perspective of the low computational cost involved while learning from large datasets. However, it may lead to loss of useful training majority instances that might have turned out to be crucial to decision-making. Oversampling the minority class is also a popular and simple balancing trick that however adds noise and involves unnecessary duplication of samples and sometimes leads to overfitting. It is possible to minimize the side-effects of sampling by incorporating intelligent procedures for pruning the majority and minority classes. This takes the form of intelligent data reduction by minimizing the loss of majority samples with class discriminatory information, and intelligent generation of new minority samples that add variety, in limits, to the existing minority population. In addition to a discussion on the strategy, the survey also enumerates the intelligent sampling procedures based on their performance, popularity, and ease of implementation. SMOTE and Borderline SMOTE are considered to be the baseline approaches by most researchers, achieving consistent classification scores in most experiments. Most of the intelligent oversampling strategies like Borderline SMOTE focus on the region near the decision boundary. However, they were found to lack a clearly defined objective function and a systematic way of generating the synthetic samples. A theoretical justification is lacking in most methods regarding the definitive gain in performance achieved by resampling. The performance depends a lot on the definitions of majority and minority classes. Multi-class imbalance correction procedures are severely lacking since most works include only binary class datasets, and if multi-class datasets were incorporated then they were first converted to binary class distributions. Also, most methods do not address the issue of drop in performance of the majority class due to addition of synthetic samples from the minority class. It was also noted that most of the researchers tend to experiment with the same set of UCI datasets such as *Pima Diabetes* and *Haberman*, and future research needs to involve a more variety in real-world datasets extracted from real-world applications. Most of these datasets are limited to few thousands of samples only. The need of the hour is to adapt the existing algorithms to Big Data applications. Especially due to the recent emergence of Internet-of-Things (IoT), the availability of large real-time datasets in large volume collected from various sensors has become a reality. The current new crop of hybrid methods that involve mutually aware conditioned treatment of the majority and minority classes, form the foundation for future research in imbalanced learning. The applications of this essential methodology

need to be successfully explored and implemented across all realms of real-world pattern recognition problems especially IoT multimodal sensor data, Big Data, and Computer Vision problems that involve thousands of image classes. New application areas need to be explored such as imbalance in text mining, real-time streaming data, web mining, bioinformatics, multimodal sensor fusion in IoT, and mining network traffic. Existing machine learning algorithms can be remodeled to incorporate class-specific balancing strategies such as the most recent work on ELMs that incorporated SMOTE. Very less information is available on how the current pre-trained deep networks that are trained on millions of images, will treat imbalanced data while performing transfer learning. Further ambiguity surrounds the deep CNNs that are trained from scratch when they are trained directly on imbalanced datasets. Additional experiments need to be performed to investigate the development of advanced deep learning modules having integrated class-specific resampling strategies.

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CONFLICT OF INTEREST

The authors declare no potential conflict of interest.

AUTHOR CONTRIBUTIONS

Seba Susan: Conceptualization; formal analysis; investigation; methodology; project administration; supervision; visualization; writing-original draft; writing-review and editing. **Amitesh Kumar:** Conceptualization; data curation; formal analysis; investigation; methodology; resources; software; validation; visualization.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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